KAIYUAN ZHANG

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EDUCATION

University of California, Los Angeles **Mar. 2026 – Present**
Ph.D. in Electrical and Computer Engineering Los Angeles, CA, USA

- Advisors: Prof. **Abeer Alwan**; Prof. **Yuan Tian**
- Overall GPA: 4.0/4.0

University of California, Los Angeles **Sep. 2024 – Mar. 2026**
M.S. in Electrical and Computer Engineering Los Angeles, CA, USA

Peking University **Sep. 2020 – Jul. 2024**
Bachelor of Science in Computer Science and Technology Beijing, China

- Overall GPA: 3.75/4.0 (Ranked 18/144)

RESEARCH INTERESTS

Speech Tokenization; Text-to-Speech Models; Post-Training of Speech-LLMs and Full-Duplex Speech Models; Speech Privacy

HONORS

- **Best Student Paper Nominee**, Interspeech 2026; **Best Paper Award**, AI4CSL Workshop at IEEE ASRU 2025; **Selected Team (Top 10 Worldwide)**, Amazon Nova AI Challenge 2026; **Outstanding Graduate**, Peking University, 2024.

EXPERIENCE

University of California, Los Angeles **Oct. 2024 – Present**
Graduate Student Researcher, advised by Prof. Abeer Alwan and Prof. Yuan Tian Los Angeles, CA

- **Speech Tokenization and Representation Learning:** Developed a discrete speech tokenizer that directly assigns SSL-derived semantic tokens to the first RVQ layer, jointly preserving acoustic fidelity and semantic information for unified speech understanding and generation, and introduced a Semantic Pre-Distillation module to reduce inference cost (ICASSP 2026 [1]). Systematically evaluated supervised and unsupervised semantic tokenization for Automatic Speech Recognition (ASR) in low-resource settings (ASRU 2025 Workshop, Best Paper Award [10]).
- **Speech Foundation Model Post-Training:** Developed post-training methods for Speech-LLMs and other speech foundation models to improve domain adaptation and personalization. Developed a domain-routed Mixture-of-Experts Speech-LLM for multi-domain ASR, reducing child-speech WER by up to 19% relative while preserving adult-speech accuracy (Interspeech 2026, Best Student Paper Nominee [2]). Developed acoustics-aware LoRA adapters with local convolutional modeling (Interspeech 2026 [9]) and an encoder-awakening strategy for effective domain adaptation of Speech-LLMs (SLT 2026 [6]). Developed a self-supervised domain adaptation method with learnable Gumbel-Softmax layer selection (Interspeech 2026 [8]). Developed a two-stage fine-tuning approach for joint child–adult role diarization and ASR (ICASSP 2027, under review [5]).
- **LLM Applications:** Released CHSER, a dataset of 200K pairs for training LLMs to correct child ASR errors, and demonstrated consistent WER reductions across domains and model architectures (Interspeech 2025 [11]). Built a pipeline combining ASR with an LLM-as-a-Judge for automated scoring and feedback in children's spoken language assessment (SLaTE 2025 [12]).
- **AI Safety, Security & Privacy:** Proposed the first black-box membership inference attack on fine-tuned TTS models at both speaker and record levels, demonstrating training-data leakage across three widely used TTS models and analyzing how speech properties affect vulnerability (NDSS 2027, under review [3]). Mitigated hallucinations in speech foundation models through uncertainty-routed activation editing at inference time (SLT 2026 [7]).

TECHNICAL SKILLS

- **Languages:** Python, C, C++, MATLAB, Bash, SQL
- **Speech & ASR Toolkits:** ESPnet, SpeechBrain, fairseq, NeMo, Kaldi, k2, torchaudio, librosa
- **Deep Learning & ML:** PyTorch, HF Transformers, NumPy, pandas, SentencePiece
- **LLM & GenAI:** RAG, QLoRA, bitsandbytes, Ollama, LLM Agent Orchestration, Agentic Coding (Claude Code, Codex)
- **Infrastructure & Tools:** Linux, Docker, Git, AWS, SLURM, Distributed Training (DDP, DeepSpeed, FSDP), Hydra, Weights & Biases

SERVICE

- **Reviewer:** Transactions on Machine Learning Research (TMLR). **Teaching Assistant:** Computer System Security; Digital Speech Processing, UCLA

PUBLICATIONS

[1] **Kaiyuan Zhang***, Mohan Shi*, Eray Eren, Natarajan Balaji Shankar, Zilai Wang, and Abeer Alwan, *STACodec: Semantic Token Assignment for Balancing Acoustic Fidelity and Semantic Information in Audio Codecs*, accepted to *ICASSP 2026*. (* equal contribution)

[2] Mohan Shi, **Kaiyuan Zhang**, Zilai Wang, Natarajan Balaji Shankar, Eray Eren, and Abeer Alwan, *Entropy-Aware Domain-Routed Mixture-of-Experts Speech-LLM Framework: A Case Study of Multi-Domain Child-Adult ASR*, accepted to *Interspeech 2026* (Best Student Paper Nominee).

[3] **Kaiyuan Zhang***, Kunlin Cai*, Zihang Xiang, Jinghuai Zhang, Abeer Alwan, Fnu Suya, and Yuan Tian, *Hearing the Whispers: Black-Box Membership Inference Attacks on Finetuned TTS Models*, under review at *NDSS 2027*. (* equal contribution)

[4] Kunlin Cai, Rui Song, Jinghuai Zhang, **Kaiyuan Zhang**, Pranav Bodapati, Alicia Yu, Fnu Suya, Mohammad Rostami, Jiaqi Ma, and Yuan Tian, *What-If World: A Causal Benchmark for General World Models in Embodied Scenarios*, under review at *ICLR 2027*.

[5] Natarajan Balaji Shankar, Zilai Wang, Mohan Shi, **Kaiyuan Zhang**, Aki Hasegawa, Farrah Galal, and Abeer Alwan, *Two-Stage Whisper Adaptation for Unified Child-Adult Role Diarization and ASR*, under review at *ICASSP 2027*.

[6] Mohan Shi, Zilai Wang, Natarajan Balaji Shankar, **Kaiyuan Zhang**, Eray Eren, and Abeer Alwan, *Encoder Awakening via Adapters: Effective Domain-Adaptive Fine-tuning of Speech-LLMs*, accepted to *SLT 2026*.

[7] Natarajan Balaji Shankar, Zilai Wang, Zihan Wang, Mohan Shi, **Kaiyuan Zhang**, and Abeer Alwan, *AURA: Uncertainty-Routed Activation Editing for Acoustic Grounding in Speech Foundation Models*, accepted to *SLT 2026*.

[8] Zilai Wang, Natarajan Balaji Shankar, Mohan Shi, **Kaiyuan Zhang**, and Abeer Alwan, *Gumbel-BEARD: Automatic Layer Selection for Self-Supervised Adaptation of Whisper in Low-Resource Domains*, accepted to *Interspeech 2026*.

[9] Natarajan Balaji Shankar, Zilai Wang, **Kaiyuan Zhang**, Mohan Shi, and Abeer Alwan, *GC-LoRA: Gated Convolutional LoRA for Parameter-Efficient Acoustic Adaptation*, accepted to *Interspeech 2026*.

[10] Mohan Shi, Natarajan Balaji Shankar, **Kaiyuan Zhang**, Zilai Wang, and Abeer Alwan, *Comparing Unsupervised and Supervised Semantic Speech Tokens: A Case Study of Child ASR*, Proc. *AI4CSL Workshop @ IEEE ASRU 2025* (Best Paper Award).

[11] Natarajan Balaji Shankar, Zilai Wang, **Kaiyuan Zhang**, Mohan Shi, and Abeer Alwan, *CHSER: A Dataset and Case Study on Generative Speech Error Correction for Child ASR*, Proc. *Interspeech 2025*.

[12] Natarajan Balaji Shankar, **Kaiyuan Zhang**, Andre Mai, Mohan Shi, Alaria Long, Julie Washington, Robin Morris, and Abeer Alwan, *Leveraging ASR and LLMs for Automated Scoring and Feedback in Children's Spoken Language Assessments*, Proc. *SLaTE 2025*.